

PLC DAY 5

Industrial Networking, HMI, SCADA, Safety & Documentation



A real PLC control cabinet connects control, networking, drives and field wiring.

Day 5 Learning Outcomes

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NETWORKING

Identify industrial network components, IP addressing and common topologies.

PROTOCOLS

Explain Modbus TCP/RTU, PROFINET and EtherNet/IP at a practical level.

HMI / SCADA

Create tags, screens, alarms, trends and operator workflows.

COMMISSIONING

Use a repeatable test and troubleshooting procedure, then document results.

From Sensor to SCADA: The Complete Signal Path

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- A sensor creates the process signal.
- Remote I/O or PLC input converts it to controller data.
- PLC logic evaluates the value and commands outputs.
- Industrial Ethernet carries selected data to HMI/SCADA.
- The operator sees status, alarms and trends—but PLC logic remains responsible for machine control.



Control cabinet: the physical home of PLC, I/O, power and network hardware.

Industrial Ethernet Hardware You Will See

- Industrial switches connect PLCs, HMIs, drives, remote I/O and engineering PCs.
- DIN-rail hardware is designed for vibration, temperature and electrical-noise environments.
- Link/activity LEDs are the first diagnostic clue during commissioning.
- Managed switches can add VLANs, redundancy, diagnostics and port monitoring.



Industrial Ethernet switch with rugged connectors and cabinet wiring.

IP Addressing Without the Confusion

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IP ADDRESS

Identifies a device on the Ethernet network.
Example: 192.168.1.20.

SUBNET MASK

Defines which addresses are local. Common lab mask: 255.255.255.0.

GATEWAY

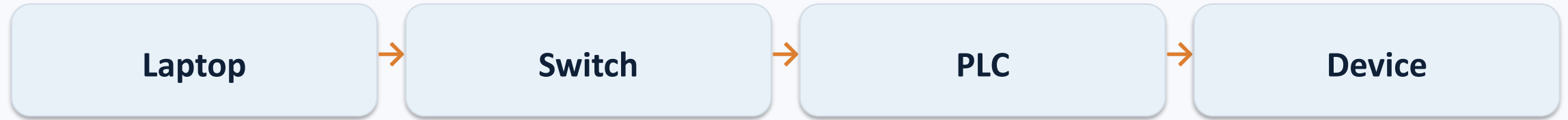
Used when communication must leave the local subnet; many isolated machine networks do not need it.

RULE

Every device needs a unique IP. Duplicate IPs create intermittent or total communication failure.

Practical Procedure: Bring a New Ethernet Device Online

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STEP 1

Connect one device at a time and check link LEDs.

STEP 2

Set compatible IP/subnet values.

STEP 3

Ping the target before configuring PLC communication.

STEP 4

Add device in engineering software and map tags.

STEP 5

Test one command and one feedback signal.

STEP 6

Record final IP, device name and firmware.

Network Topologies: What Changes in the Field

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STAR

Most devices connect to a central switch.
Easy to troubleshoot and expand.

LINE / DAISY CHAIN

Devices connect sequentially. Saves cabling
but a break can affect downstream nodes.

RING

Used with redundancy protocols to maintain
communication after one cable/path failure.

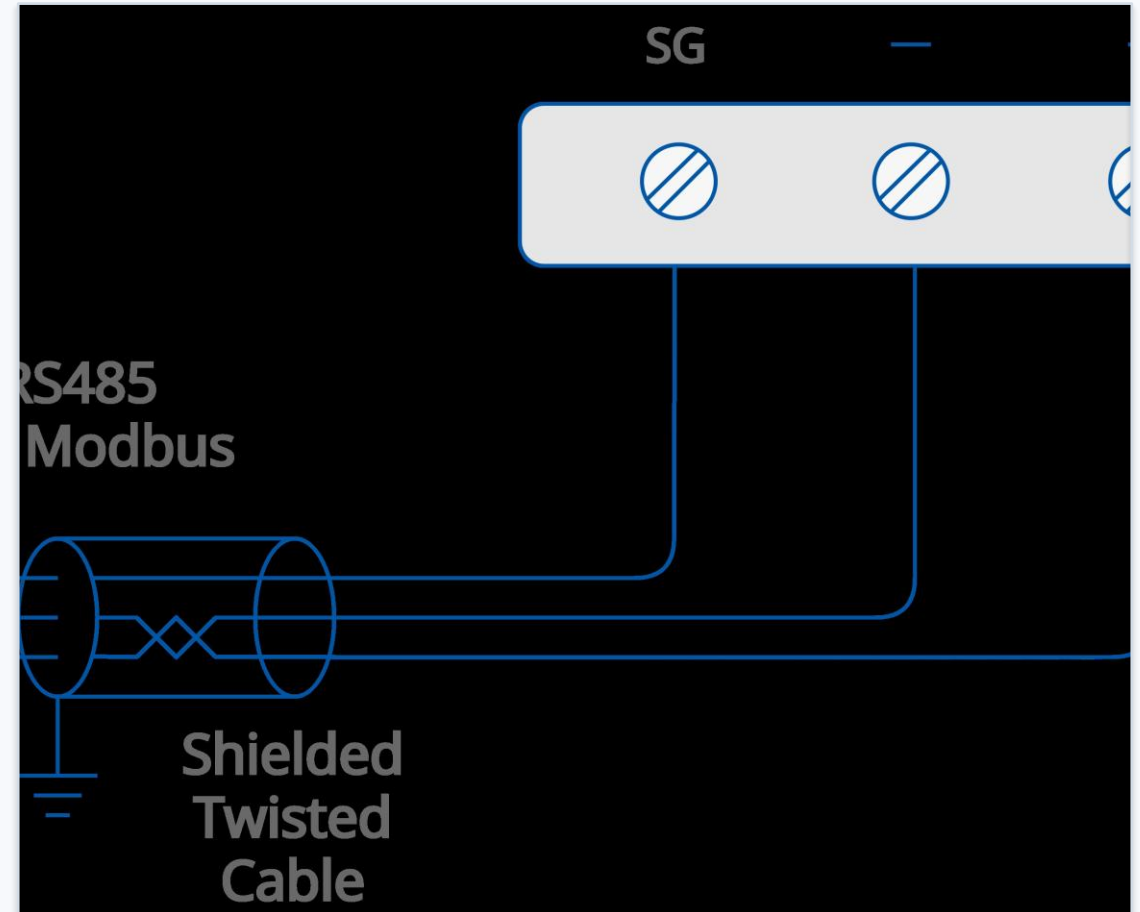
SEGMENTED NETWORK

Separate machine, supervisory and
enterprise traffic to improve control and
security.

Modbus RTU: What the Wiring Really Looks Like

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- Modbus RTU commonly runs over RS-485 differential serial communication.
- Use a daisy-chain bus with correct A/B polarity and appropriate termination.
- Each slave/server must have a unique node address.
- All devices must match baud rate, parity and stop bits.
- If nothing communicates: check wiring and serial settings before touching PLC logic.



RS-485 / Modbus RTU wiring concept. Always confirm A/B naming in the device manual.

Modbus RTU Message Concept

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MASTER / CLIENT

Initiates a request: read or write specific data.

SLAVE / SERVER

Responds only when addressed.

FUNCTION CODE

Defines the action, such as reading coils or holding registers.

CRC

Error-checking field used to detect corrupted serial frames.

Modbus Data Areas Students Must Recognize

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COILS

1-bit read/write values: commands, enables, discrete outputs.

DISCRETE INPUTS

1-bit read-only values: status or digital input states.

INPUT REGISTERS

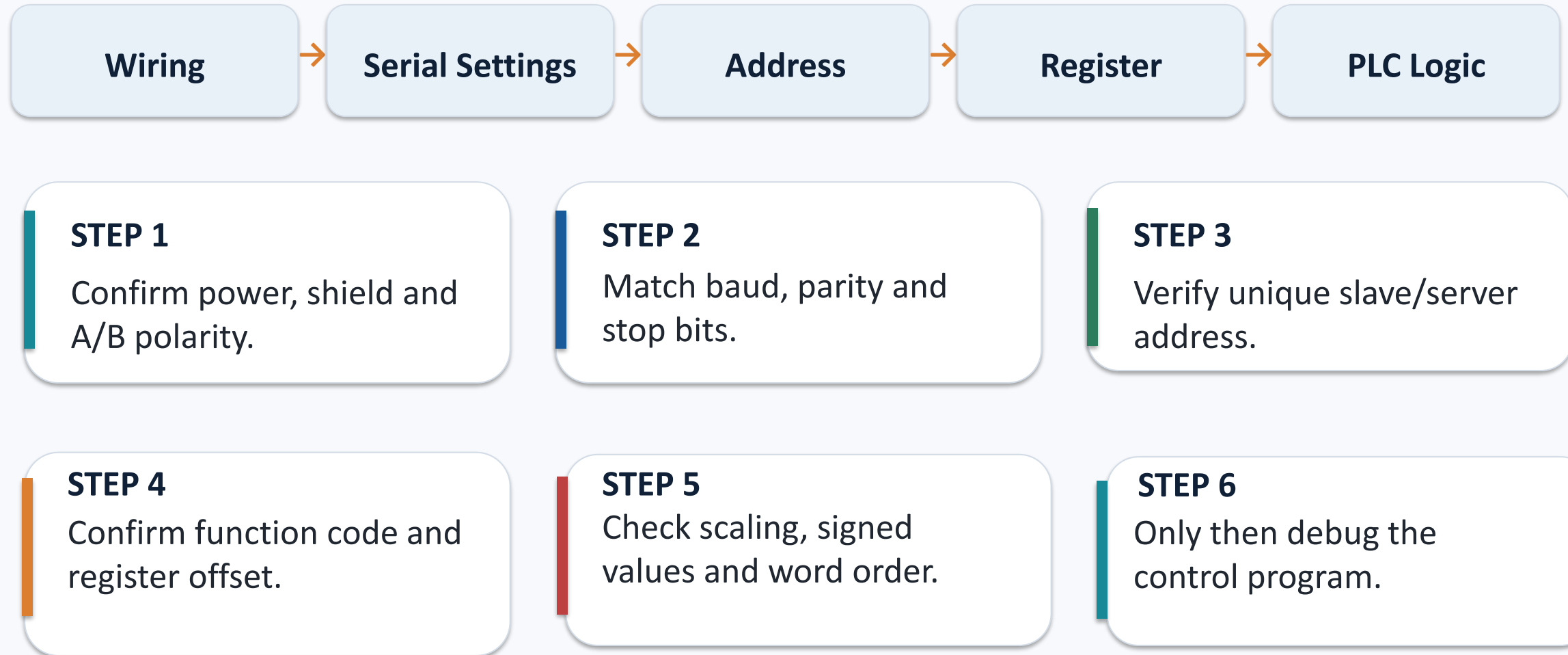
16-bit read-only process measurements.

HOLDING REGISTERS

16-bit read/write values: setpoints, parameters and process data.

Modbus RTU Troubleshooting Order

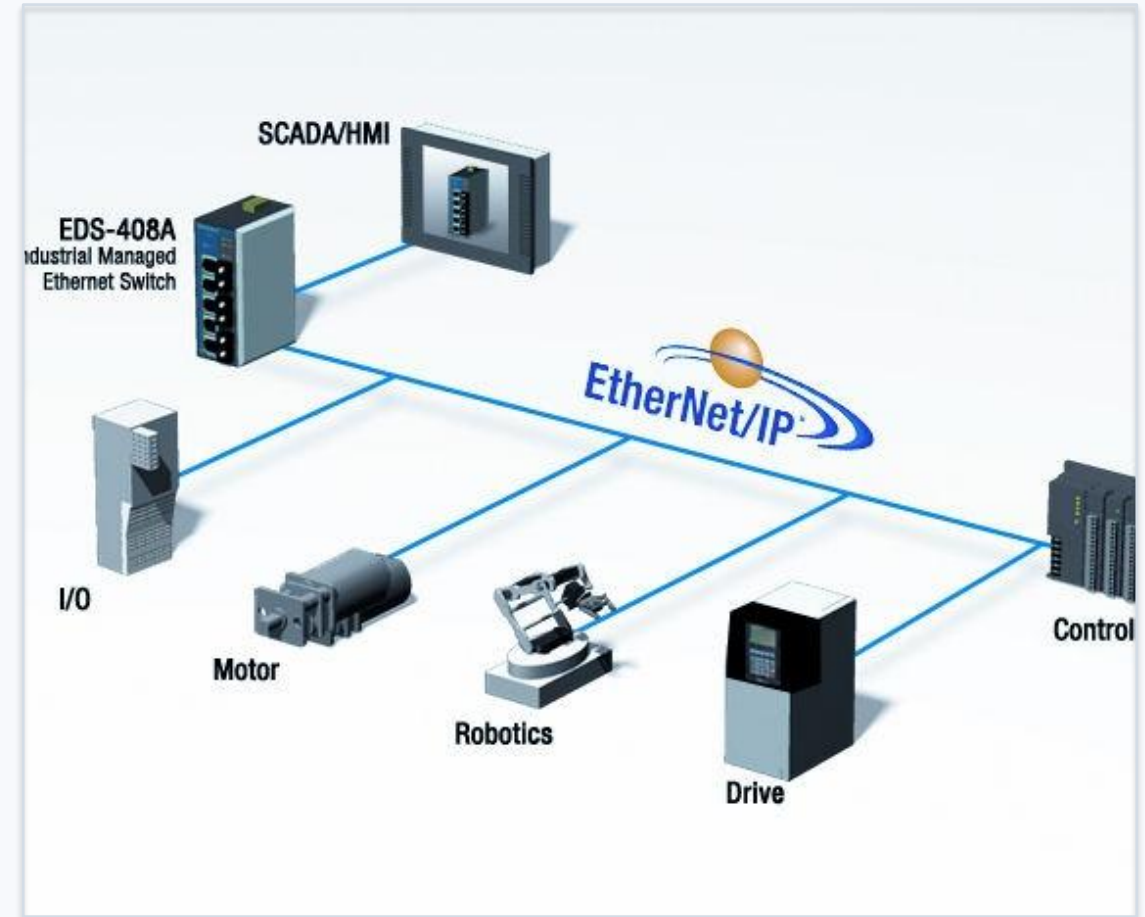
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Modbus TCP: Same Data Model, Ethernet Transport

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- Modbus TCP carries Modbus requests through Ethernet/TCP rather than serial RS-485.
- Typical communication uses IP addressing and TCP port 502.
- The register model remains familiar: coils, inputs and registers.
- Practical benefit: easier integration with Ethernet switches, HMIs, gateways and SCADA systems.



Example industrial Ethernet automation network.
Modbus TCP uses the same physical Ethernet infrastructure.

16-bit vs 32-bit Data: Common Commissioning Trap

16-BIT REGISTER

One Modbus register = 16 bits. It may represent UINT, INT, status bits or part of a larger value.

32-BIT VALUE

REAL/DINT often uses two consecutive registers.

WORD ORDER

Two devices may store high/low words in different order.

SYMPTOM

Wrong word order produces values that look random even though communication is healthy.

Scaling Raw Network Values

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RAW VALUE

Device may send 0–10000 instead of engineering units.

NORMALIZE

Convert raw range to 0.0–1.0.

SCALE

Map normalized value to engineering range, e.g. 0–50 Hz.

VERIFY

Compare PLC value with the device display at low, mid and high points.

PROFINET: Practical View

- PROFINET is an industrial Ethernet system widely used for controller-to-I/O and drive communication.
- Devices are configured with network identity and cyclic process data.
- Engineering software maps device I/O into PLC tags/addresses.
- Commissioning focuses on device name/IP, hardware configuration, module order and diagnostics.



Industrial Ethernet hardware used to interconnect controllers and distributed devices.

PROFINET Commissioning Checklist

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1 • DISCOVER

Find reachable devices and confirm physical network.

2 • IDENTIFY

Assign/verify device name and IP settings.

3 • CONFIGURE

Match hardware modules and I/O layout to the real device.

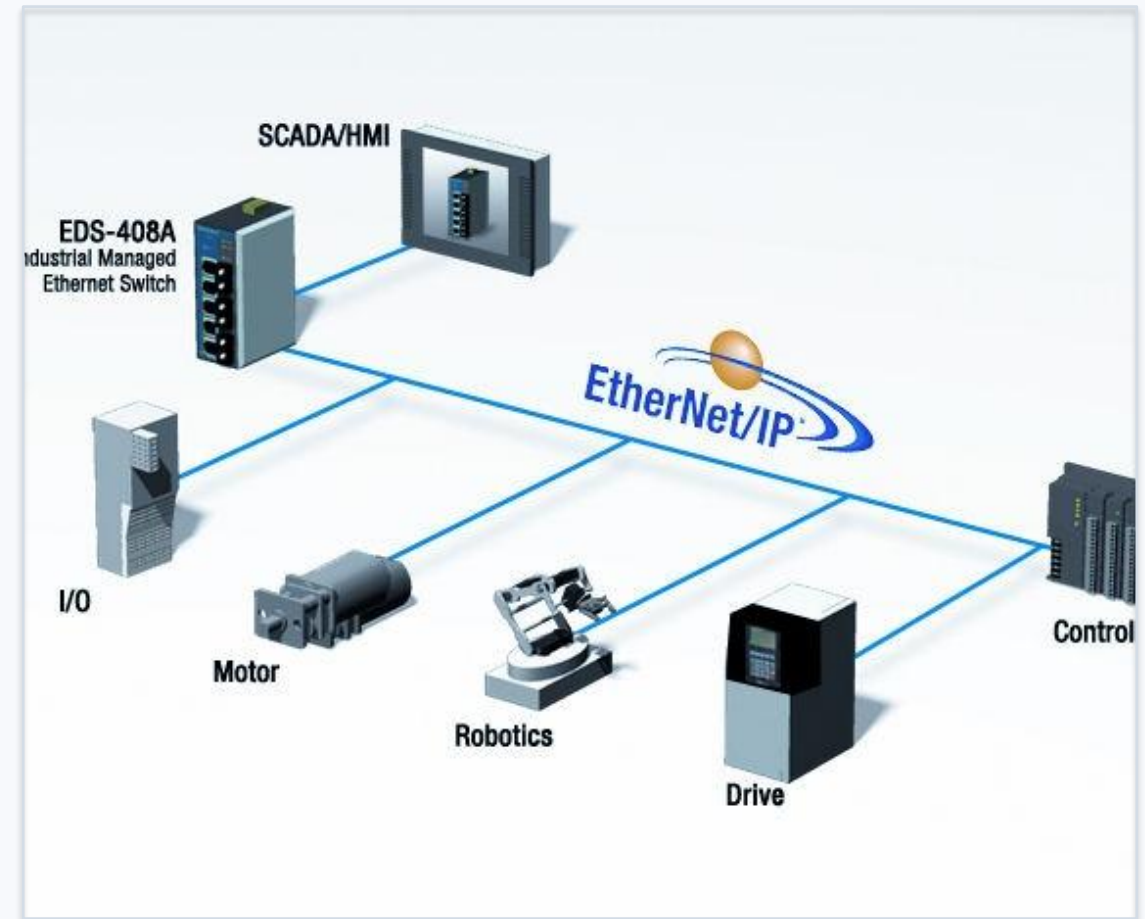
4 • TEST

Download, go online, verify module status and test one I/O point at a time.

EtherNet/IP: Practical View

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- EtherNet/IP uses Ethernet with the Common Industrial Protocol (CIP).
- Controllers exchange cyclic I/O with drives, remote I/O and other devices.
- Device configuration may include input/output assembly instances and data sizes.
- Troubleshooting begins with IP/subnet, physical link and connection status before checking mapped tags.



EtherNet/IP network example connecting controllers, drives, robots and supervisory devices.

PROFINET vs EtherNet/IP vs Modbus TCP

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PROFINET

Strong device-oriented engineering and cyclic I/O; common in Siemens ecosystems.

ETHERNET/IP

CIP-based cyclic I/O and explicit messaging; common in Rockwell ecosystems.

MODBUS TCP

Simple register-based client/server communication; broadly supported.

SELECTION

Real projects choose based on controller ecosystem, device support, diagnostics and plant standards.

VFD Networking: What Data Actually Moves

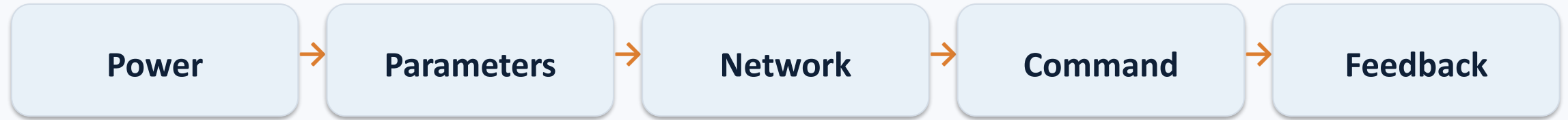
- PLC → VFD: run/stop, direction, speed reference and reset command.
- VFD → PLC: ready, running, actual speed/frequency, current and fault code.
- Never assume communication alone makes the machine safe—hardwired safety circuits may still be required.
- Before test: verify motor data, command source, reference source and acceleration/deceleration settings.



Industrial variable-frequency drives used to control AC motor speed.

VFD Commissioning Procedure

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STEP 1

Verify safe wiring and motor nameplate data.

STEP 2

Set command/reference source to the intended control mode.

STEP 3

Configure address/IP and communication parameters.

STEP 4

Test low-speed run before full-speed operation.

STEP 5

Compare commanded vs actual speed/current.

STEP 6

Force a controlled fault and verify PLC/HMI response.

HMI: The Operator's Window into the Machine

- The HMI should show machine state, process values, commands, alarms and diagnostics clearly.
- The operator should understand what is running and what is preventing a start within seconds.
- Use meaningful equipment names and engineering units.
- PLC permissives and interlocks must remain in PLC logic; the HMI requests actions, it does not replace safety logic.



Industrial touchscreen HMI mounted on production equipment.

HMI Tag Architecture

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PLC TAG

Source of truth in the controller: Motor_Run, Tank_Level, Alarm_HighTemp.

HMI TAG

Linked to PLC variable or communication address.

DISPLAY OBJECT

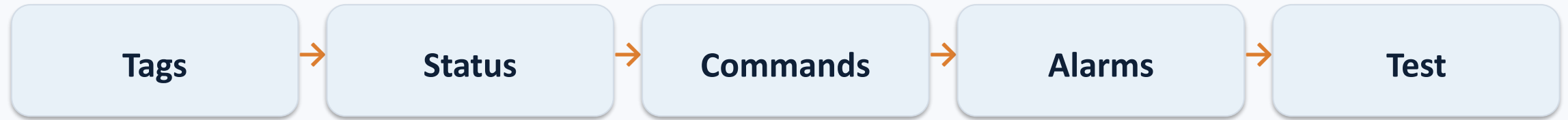
Lamp, numeric display, bar graph or status text reads the tag.

COMMAND OBJECT

Button or entry field writes a request/setpoint back to the PLC.

Procedure: Build One HMI Screen Correctly

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STEP 1

Create and verify PLC tags first.

STEP 2

Show feedback/status before adding buttons.

STEP 3

Add Start/Stop/Reset and setpoint entry.

STEP 4

Add units, limits and alarm banner.

STEP 5

Test every command and every feedback.

STEP 6

Finally test communication loss and restart behavior.

Good HMI vs Bad HMI

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GOOD

Consistent navigation, neutral background, clear status, units, limited color and obvious abnormal conditions.

BAD

Rainbow colors, tiny text, decorative animation, unexplained abbreviations and no clear alarm priority.

COLOR RULE

Reserve strong colors for abnormal or actionable conditions.

DESIGN TEST

A new operator should understand the machine state without asking what every color means.

SCADA: Beyond a Single Machine

- SCADA supervises multiple PLCs, machines or process areas from centralized workstations.
- It collects live values, alarms, events and historical trends.
- Operators can issue supervisory commands while local PLCs continue real-time control.
- Engineering teams use SCADA history to investigate downtime, quality and process behavior.



Real SCADA control-room environment with multiple process and alarm displays.

FIELD LEVEL

Sensors, actuators, drives and instruments.

CONTROL LEVEL

PLCs / RTUs execute real-time control.

SUPERVISORY LEVEL

HMI/SCADA servers and operator stations visualize and supervise.

HISTORY / REPORTING

Historian, event database and reports preserve process information.

Alarm Philosophy: More Than a Red Lamp

CONDITION

Define exactly what makes the alarm active.

PRIORITY

Use severity based on consequence and required response.

MESSAGE

Tell the operator what happened and where—not just 'FAULT 12'.

RESPONSE

Define acknowledgement, cause removal and reset behavior.

Alarm Lifecycle Students Should Test



STEP 1

Start from a healthy process.

STEP 2

Create one known abnormal condition.

STEP 3

Verify alarm text, time and priority.

STEP 4

Acknowledge: active fault must remain visible.

STEP 5

Remove the physical/process cause.

STEP 6

Reset only when restart conditions are safe.

Alarm Hysteresis and Delay

HYSTERESIS

Prevents an analog alarm from repeatedly toggling near its threshold.

ON-DELAY

Requires the condition to remain abnormal for a defined time before alarming.

OFF-DELAY

Can prevent rapid clearing when a signal briefly returns normal.

EXAMPLE

High temperature alarm at 80°C, clear below 77°C after stable recovery.

Event Logging vs Alarm Logging

EVENT

Records an occurrence: mode changed, recipe loaded, operator login, motor started.

ALARM

Represents an abnormal condition requiring awareness or action.

WHY LOG

Timeline reconstruction helps diagnose failures and operator/process sequence.

PRACTICAL

Compare event timestamps with trends to understand what happened before a trip.

Trends: Turning Values into a Story

LIVE TREND

Shows current behavior while commissioning or tuning.

HISTORICAL TREND

Stores data for later investigation.

PEN SELECTION

Trend only variables that answer a question: setpoint, PV, output, current, pressure, etc.

TIME BASE

Choose seconds for fast events; minutes/hours for slow processes.

Control Room Thinking: What Operators Need

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- Overview screens answer: Is the plant healthy?
- Detail screens answer: What exactly is wrong here?
- Alarm displays answer: What requires action first?
- Trend displays answer: How did the process reach this condition?
- Good SCADA design reduces searching and supports faster, safer decisions.



SCADA workstations combine overview, detail, alarm and trend information.

Communication-Loss Strategy

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DETECT

Use connection status, heartbeat or watchdog logic.

ANNUNCIATE

Show a clear communication alarm and affected device.

SAFE RESPONSE

Define what outputs/setpoints should do if communication disappears.

RECOVERY

Decide whether reconnection resumes automatically or requires operator confirmation.

CONTROL ≠ SAFETY

A normal PLC/HMI command is not automatically a safety function.

E-STOP

Emergency stop circuits require appropriate safety architecture and hardware.

SAFE STATE

Define what the machine must do when power, communication or a critical sensor is lost.

TESTING

Safety validation must be planned and documented—not assumed from normal operation.

Inside a Control Cabinet: What to Document

- PLC model and module order.
- Power supplies, breakers, relays and terminal blocks.
- Network switch ports and connected device names.
- VFD model, motor assignment and communication settings.
- Wire numbers, terminal numbers and field device references.



A well-organized cabinet makes commissioning and maintenance much easier.

NETWORK MAP

Device name, IP address, protocol, switch port and physical location.

I/O LIST

PLC address/tag, field device, signal type, range and engineering unit.

ALARM LIST

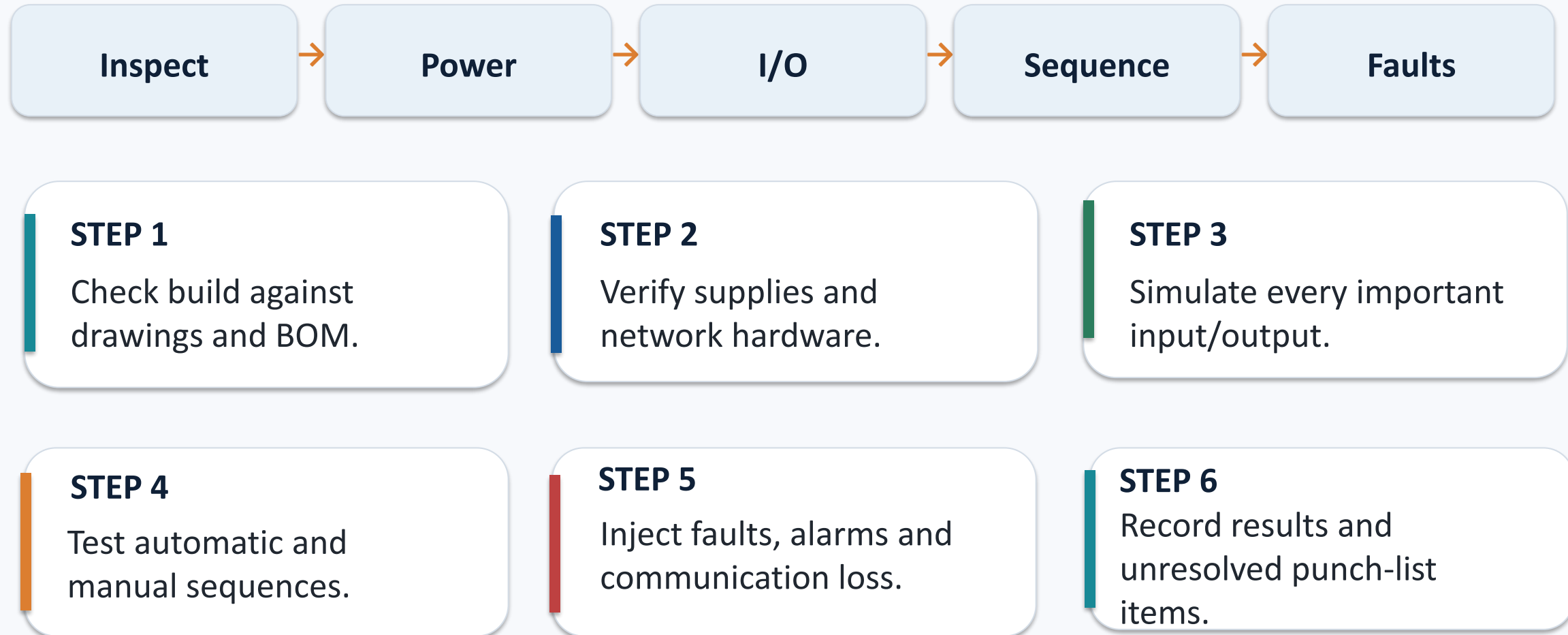
Alarm name, trigger, priority, delay, reset and operator response.

BACKUP RECORD

PLC/HMI/drive versions, firmware, parameter backups and change history.

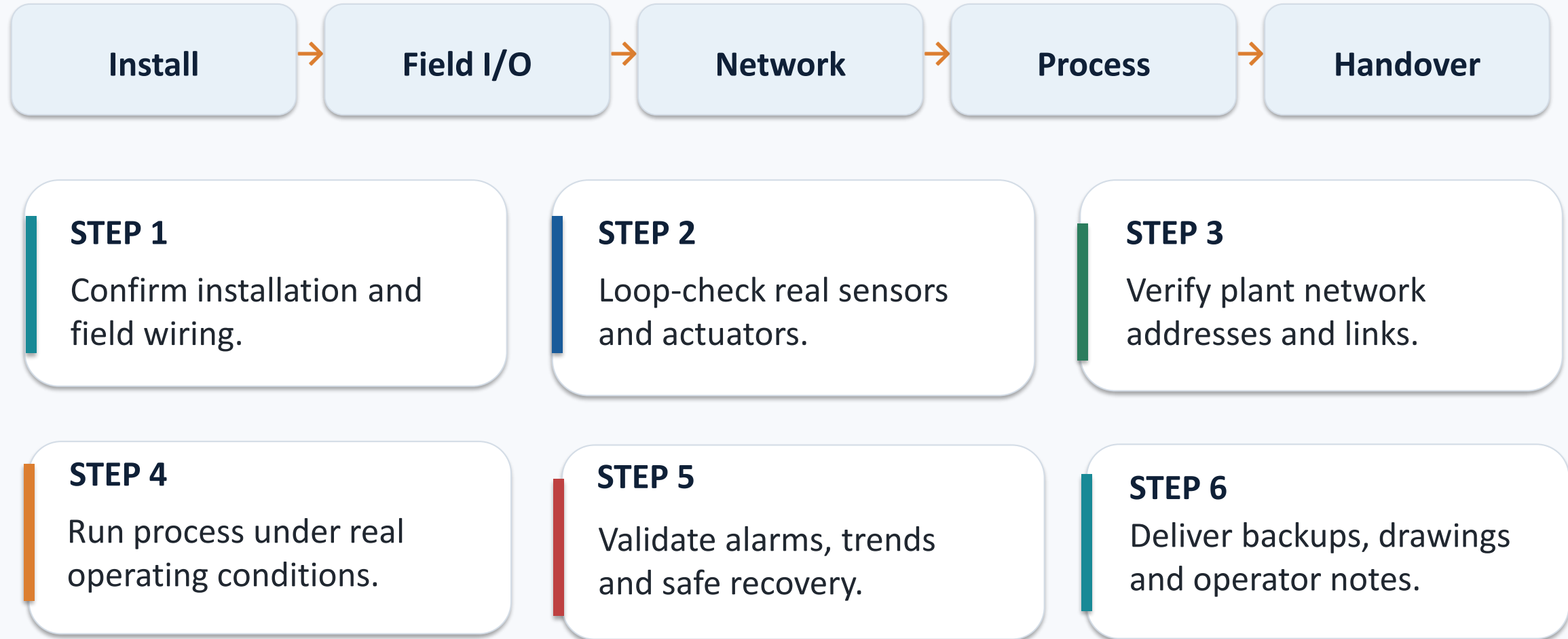
FAT: Factory Acceptance Test

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SAT: Site Acceptance Test

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Day 5 Practical Student Challenge

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TASK 1

Connect PLC, HMI and one networked device through an Ethernet switch.

TASK 2

Create status, command, setpoint, alarm and trend tags.

TASK 3

Run a small motor/conveyor/pump sequence and display live feedback.

TASK 4

Disconnect the network, create a fault, recover safely and document what happened.

DAY 5 TAKEAWAY

A working industrial system is not only PLC logic.

It is the combination of correct networking, device configuration, HMI/SCADA design, alarm handling, safe failure behavior, testing and documentation.

NEXT: integrate the Day 5 concepts into the sorting & packaging capstone.



Industrial networking is the communication backbone of modern automation.